

PHAT CAO

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EDUCATION

George Mason University

Master of Science in Computer Science - GPA: 3.67/4.00

Expected May 2028

Fairfax, VA

Virginia Tech

Bachelor of Science in Computer Science - GPA: 3.60/4.00

Aug 2021 - May 2025

Blacksburg, VA

TECHNICAL SKILLS

Languages: Java • Python • JavaScript • TypeScript • SQL • C • HTML • CSS

Frameworks & Libraries: React • Vue • Node.js • Flask • Alpine.js • Tailwind CSS • Playwright • JUnit

Technologies: Git • Docker • Docker Compose • Maven • Apache Kafka • WebSockets • REST APIs • Linux/Unix • CI/CD

Databases: MySQL • MongoDB

PROFESSIONAL EXPERIENCE

Handshake AI Fellowship

May 2026 - Aug 2026

AI Technical Contributor (Contract)

Remote

- Developed Python solutions for data processing, database recovery, and validation challenges using SQLite, JSON, and automation pipelines while ensuring correctness through comprehensive testing.
- Evaluated multimodal AI model outputs across image, video, and document understanding tasks, identifying reasoning errors, instruction-following failures, and data quality issues.
- Applied detailed evaluation guidelines to produce high-quality annotations and technical justifications, contributing to AI model benchmarking and quality assurance.

Peraton

Aug 2024 - Dec 2024

Software Engineer Co-op

Blacksburg, VA

- Enhanced reliability of a cyber-threat intelligence platform by resolving production backend issues, fixing frontend API integration bugs, and standardizing Docker Compose development environments.
- Migrated an enterprise Apache Kafka cluster from ZooKeeper to KRaft, simplifying CI/CD infrastructure and eliminating the ZooKeeper dependency across distributed environments.
- Restored build stability and resolved 100+ Maven dependency conflicts, reducing security vulnerabilities across backend microservices.

Rural Trash Collection (Independent Study)

Jan 2024 - Dec 2024

Virginia Tech IDPro

Blacksburg, VA

- Led a software team building an autonomous trash-collection robot, coordinating development, C++ integration, and bi-weekly stakeholder reviews with VDOT.
- Developed C++ software on a Raspberry Pi to process LiDAR data for obstacle detection, integrating sensor readings into the robot's navigation logic.
- Designed a reusable training pipeline for future ML object detection models, reducing onboarding time for new contributors.

Card Isle

Sept 2023 - Nov 2023

Front-End Developer Intern

Blacksburg, VA

- Improved visual consistency across 10+ front-end components by standardizing styling patterns with Tailwind CSS, reducing UI discrepancies across customer-facing pages.
- Achieved a 95%+ pass rate across automated E2E test suites by authoring Playwright scripts simulating user flows on Chrome, Firefox, and Safari, preventing cross-browser regressions prior to code review.

TECHNICAL PROJECTS

The Odd One

Aug 2026 - Present

Vue.js • TypeScript • Node.js • WebSockets • Pinia • Vercel

Blacksburg, VA

- Built a real-time multiplayer social-deduction web application using Vue and TypeScript by supporting room creation, player synchronization, configurable game settings, and multiple game modes.
- Designed a WebSocket-based client-server architecture to synchronize game phases, player actions, timers, voting, and results across connected clients.

Traffic DashBoard Visualization (Senior Capstone Project)

Jan 2025 - May 2025

React.js • Chart.js • Node.js • Express.js • Docker • MySQL

- Led backend development for a five-person team by designing REST APIs, MySQL database schema, and backend architecture.
- Redesigned the data pipeline by replacing large JSON parsing with normalized MySQL storage and REST API queries.
- Reduced data load latency by 50% by implementing backend API queries instead of loading large in-memory JSON structures.
- Containerized the full-stack application with Docker Compose to streamline setup and deployment.